WEEK 10 – TUTORIAL

Objectives:

This tutorial aims to help students understand how **project management knowledge areas align with the project lifecycle and the critical role of integration management in IT projects**. Students will apply their knowledge by mapping active knowledge areas to specific lifecycle stages, analysing a real-world IT scenario involving changing requirements and integration challenges, identifying relevant integration management processes, proposing strategies to manage work and coordinate stakeholders, and evaluating risks. Additionally, students will reflect on the difficulties of integration processes and the potential impact on knowledge areas when integration is weak, reinforcing practical and analytical skills for managing complex IT projects.

Tutorial Activities:

- Knowledge Area Mapping
- Scenario Analysis

Learning outcomes

By the end of this tutorial, students will be able to:

- **LO1**. Understand each of the core project management knowledge areas
- **LO4**. Develop basic proficiency in project management tools and practices

Part A – Knowledge Area Mapping

Each team is given 1 project lifecycle stage. Each team has to list 2-3 knowledge areas that are most active in that stage and briefly explain why.

Project Lifecycle Stage	***Most Active Knowledge Areas***	***Explanation***
Initiation	*Stakeholder Management, Integration Management, Communication Management*	*Stakeholder input is critical to authorize the project. Integration ensures all objectives are aligned; communication ensures stakeholders understand project goals.*
Planning	*Scope Management, Schedule Management, Cost Management*	*Defining deliverables, creating timelines, and budgeting are key during planning. Integration ensures all subsidiary plans are consolidated.*
Execution	*Resource Management, Quality Management, Communication Management*	*Teams execute tasks; managing resources and ensuring quality are critical. Communication keeps teams and stakeholders aligned.*
Monitoring & Controlling	*Risk Management, Scope Management, Integration Management*	*Monitoring performance, controlling changes, and mitigating risks are central. Integration ensures corrective actions align with overall project goals.*
Closing	*Integration Management, Communication Management, Procurement Management*	*Formal closure of project deliverables, archiving documentation, releasing resources, and communicating outcomes. Integration ensures smooth handover and closure.*

Part B – Scenario analysis

Your team is tasked with implementing a university online learning platform. Midway through the project, stakeholders request a new virtual lab feature, and the database integration shows performance issues.

- Identify which integration management processes would be triggered.
  Perform Integrated Change Control → Evaluate the impact of the new virtual lab feature on scope, schedule, cost, and resources.
  Direct & Manage Project Work → Adjust work assignments to incorporate new features and fix database issues.
  Monitor & Control Project Work → Track progress of integration fixes and new feature development.
  Manage Project Knowledge → Document lessons learned from database performance issues for future projects.
- Suggest how you would direct and manage project work to handle the new feature and database issue.
  Reprioritize tasks to address database performance bottlenecks first.
  Assign dedicated sub-teams for virtual lab development and integration testing.
  Implement daily stand-ups or scrum meetings to monitor progress and resolve blockers.
  Update the project schedule and resource allocation to accommodate the new feature.
- Propose at least two tools or techniques you would use to coordinate teams and stakeholders.
  Project Management Tools: Jira, MS Project, or Trello for task tracking and progress monitoring.
  Collaboration & Communication Tools: Slack, Microsoft Teams, or Confluence for documentation, updates, and stakeholder communication.
  Optional: Version control (Git) to manage code changes and integration efficiently.
- List one risk introduced by the new feature and how you would mitigate it.
  Risk: Scope creep and delayed delivery due to added complexity.
  Mitigation: Perform a thorough impact analysis, update project plan and timeline, communicate trade-offs with stakeholders, and approve changes via integrated change control.
- Which Integration Management process do you think is most difficult in IT projects?
  Perform Integrated Change Control is often most challenging because:
    IT projects frequently experience scope changes due to evolving requirements.
    Changes affect multiple knowledge areas (schedule, cost, resources, quality).
    Requires coordination across multiple teams and stakeholders.
- Which knowledge area is most impacted if integration is weak?
  Scope Management is most impacted:
    Weak integration can lead to uncontrolled scope changes, scope creep, and misalignment with project objectives.
  Also Affected: Schedule, Cost, and Quality, as they are interdependent.